

Transport Management System - ER Diagram & Project Roadmap

Database Structure Requirements

1. Justification for Tracking Table

The 'Tracking' table is vital for maintaining a real-time event log of a shipment's journey (Lorry Receipt / Bilty).

Why is this required?

- Auditing & Transparency: It logs every status state change ('In-Transit', 'At Checkpoint', 'Delivered') over time, providing a clear history rather than just the current status.
- Customer Visibility: End customers and internal administrators need to track exact vehicle locations to trust the service and manage expectations.
- Operational Bottlenecks: By analyzing the timestamps of the tracking logs, management can measure delays, find inefficient routes, and optimize travel plans.
- Disruption Management: If a vehicle breaks down, the tracking table allows immediate logging of exceptions so that all stakeholders are automatically notified.

2. Table Structure: EWayBill (17 Columns)

Stores official Electronic Way Bill document details compliant with GST requirements.

Column Name	Data Type	Null / Not Null	Data Usage & Null Handling
1. EWayBillNo (PK)	VARCHAR(20)	NOT NULL	Primary Key. Filled mandatory via govt portal.
2. EWayBillDate	DATETIME	NOT NULL	Generation timestamp. Mandatory.
3. LrNumber (FK)	VARCHAR(30)	NOT NULL	Links to LrBilty table. Mandatory.
4. GenByGSTIN	VARCHAR(15)	NOT NULL	GSTIN of generator. Mandatory.
5. SupplierGSTIN	VARCHAR(15)	NOT NULL	Supplier GST. Mandatory.
6. RecipientGSTIN	VARCHAR(15)	NOT NULL	Recipient/Buyer GST. Mandatory.
7. DispatchFrom	VARCHAR(200)	NOT NULL	Origin location/pincode. Mandatory.
8. ShipTo	VARCHAR(200)	NOT NULL	Destination location/pincode. Mandatory.
9. Distance	INT	NOT NULL	Delivery distance in Km. Mandatory.
10. TotalInvoiceVal	DECIMAL(10,2)	NOT NULL	Total value of goods. Mandatory.
11. CustomDutyVal	DECIMAL(10,2)	NULL	Leave empty if No Custom Duty. Data if applicable.
12. CGSTAmount	DECIMAL(10,2)	NULL	Put 0 or leave empty if exempt, else proper value.
13. SGSTAmount	DECIMAL(10,2)	NULL	Put 0 or leave empty if exempt, else proper value.
14. IGSTAmount	DECIMAL(10,2)	NULL	Put 0 or leave empty if exempt, else proper value.
15. VehicleNumber	VARCHAR(20)	NOT NULL	Registered Truck Number. Mandatory.
16. TransporterID	VARCHAR(50)	NULL	Transporter ID. Leave empty if transported internally.
17. ValidUpto	DATETIME	NOT NULL	Expiration time of E-Way Bill. Mandatory.

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3. Table Structure: LrBilty (17 Columns)

Acts as the core Lorry Receipt/Bilty for the primary freight transport contract.

Column Name	Data Type	Null / Not Null	Data Usage & Null Handling
1. LrNumber (PK)	VARCHAR(30)	NOT NULL	Primary Key (e.g., LR-1001). Mandatory.
2. LrDate	DATE	NOT NULL	Date the receipt was created. Mandatory.
3. ConsignorID	INT	NOT NULL	Sender ID linked to Customers Table. Mandatory.
4. ConsigneeID	INT	NOT NULL	Receiver ID linked to Customers Table. Mandatory.
5. Origin	VARCHAR(100)	NOT NULL	City/Branch trip origin. Mandatory.
6. Destination	VARCHAR(100)	NOT NULL	City/Branch trip destination. Mandatory.
7. MaterialDesc	TEXT	NOT NULL	Description of the products. Mandatory.
8. TotalWeight	DECIMAL(10,2)	NOT NULL	Total volume/dead weight. Mandatory.
9. TotalPackages	INT	NOT NULL	Count of boxes/cartons. Mandatory.
10. FreightAmount	DECIMAL(10,2)	NOT NULL	Total transport fare. Mandatory.
11. AdvanceAmount	DECIMAL(10,2)	NULL	Pre-paid transport amount. Empty if None.
12. BalanceAmount	DECIMAL(10,2)	NULL	Pending amount. Empty if 0 balance.
13. VehicleID	INT	NULL	Link to Vehicle table. Empty until designated.
14. DriverID	INT	NULL	Link to Driver table. Empty until designated.
15. EWayBillNo	VARCHAR(20)	NULL	Empty until Govt Bill generated.
16. DeliveryDate	DATETIME	NULL	Empty while in transit. Filled only upon delivery.
17. Status	VARCHAR(30)	NOT NULL	Set to "Created", "In Transit", or "Delivered".

4. Additional Requirements for the ER Diagram

To complete the full Database Structure and allow correct connectivity (Foreign Keys), the following additional tables should be mapped to the core system in your ER diagram:

1. Customers Table (Important)

- Connection: Connects to LrBilty (ConsignorID, ConsigneeID).
- Data needed: Customer Name, Billing Address, GSTIN, Phone.

2. Vehicles Table

- Connection: Connects to LrBilty (VehicleID) and Tracking Table.
- Data needed: Vehicle Registration Number, Owner Name, RC Expiry.

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3. Drivers Table

- Connection: Connects to LrBilty (DriverID).
- Data needed: Driver Name, License Number, Contact Phone.

4. Users / Staff Table

- Connection: Connects to Tracking (UpdatedBy) and LrBilty (CreatedBy).
- Data needed: Admin Name, Role, Login Credentials.

5. Invoices / Payments Table

- Connection: Connects to LrBilty.
- Data needed: Transactions for Advanced vs. Balanced Freight amounts.